A

Project Report

on

A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening

submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY DEGREE

SESSION 2025-26

in

COMPUTER SCIENCE ENGINEERING (ARTIFICIAL INTELLIGENCE)

By

LAKSHYA BHARGAVA (2200291520104)
PRAFUL SRIVASTAVA (2200291520127)
PARSHAV JAIN (2200291520124)
PRATHAM ANTIWAL (2200291520132)

Under the supervision of

Mr. Rishabh Sachan

KIET Group of Institutions, Ghaziabad

Affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

(Formerly UPTU)

May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Name:- LAKSHYA BHARGAVA

Roll No.:- 2200291520104

Signature:-

Name:- PRAFUL SRIVASTAVA

Roll No.:- 2200291520127

Signature:-

Name:- PARSHAV JAIN

Roll No.:- 2200291520124

Signature:-

Name:- PRATHAM ANTIWAL

Roll No.:- 2200291520132

Signature:-

Date:

CERTIFICATE

This is to certify that Project Report entitled "A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening" which is submitted by LAKSHYA BHARGAVA, PRAFUL SRIVASTAVA, PARSHAV JAIN, PRATHAM ANTIWAL in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Mr. Rishabh Sachan **Dr. Rekha Kashyap**

(Associate Professor) **Dean – CSE (AI/AI&ML)**

Date:

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to Mr. Rishabh Sachan, Department of CSE(AI), KIET, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr. Rekha Kashyap, Dean of the Department of Computer Science & Engineering (AI/AI&ML), KIET, Ghaziabad, for her full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project.

Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

Date:

Name:- LAKSHYA BHARGAVA

Roll No.:- 2200291520104

Signature:-

Name:- PRAFUL SRIVASTAVA

Roll No.:- 2200291520127

Signature:-

Name:- PARSHAV JAIN

Roll No.:- 2200291520124

Signature:-

Name:- PRATHAM ANTIWAL

Roll No.:- 2200291520132

Signature:-

ABSTRACT

Thyroid disorders constitute a common but still frequently overlooked health problem. The process of discovering thyroid disorders is still very complicated and is especially challenging in rural healthcare where there are limited resources, as specialists' consultations and lab tests are often unavailable. Moreover, announcing the disorder early and correctly is imperative since it will lead to improved patient outcomes and less burden on the health system. This article proposes a two- stage, multimodal machine learning framework for the early screening of thyroid disorders. It integrates three modalities: visual examination of neck images, symptom-related questionnaires, and optional thyroid hormone profiles (TSH, T3, T4). In Stage 1, a lightweight convolutional neural network (CNN) scrutinizes the clinical neck image to predict the likelihood of goiter, hypothyroidism, hyperthyroidism, and thyroid nodules, while a corresponding symptom-scoring module also augments the accuracy of the prediction. In this way, the cases that are sure enough are marked immediately while the borderline predictions move to Stage 2. During the second stage, the structured hormone test values are combined with the symptom data in order to clear the unclear or misleading hormone patterns by means of a calibrated decision model. The first test on a pilot dataset reveals that the multimodal fusion approach significantly increases the stability of predicting when compared to either image-only or hormone-only techniques. The hybrid methodology has elevated the overall accuracy and the resilience of the system, thus, addressing the drawbacks of laboratory testing. The system has been developed as a user-friendly, point-of-care screening tool accessible for all cases and not as a diagnostic substitute, thus, targeting early detection in deprived regions. This framework has set a landmark for the realization of community-level thyroid screening at scale and it also opens up opportunities for integration with mobile health platforms, telemedicine, and public health screening programs.

TABLE OF CONTENTS

Page No.

DECLARATION 2

CERTIFICATE 3

ACKNOWLEDGEMENT 4

ABSTRACT 6

LIST OF FIGURES 10

LIST OF TABLES 11

LIST OF ABBREVIATIONS 12

CHAPTER 1 (INTRODUCTION) 13

1.1. Introduction 13

1.2. Project Description 15

CHAPTER 2 (LITERATURE REVIEW) 17

2.1. Introduction 17

2.2. Machine Learning in Medical Diagnosis 18

2.3. Thyroid Disease Detection Using Hormone-Based Models 19

2.4. Deep Learning for Medical Image Analysis 20

2.5. Symptom-Based Clinical Scoring Systems 20

2.6. Multimodal Machine Learning Systems 21

2.7. Research Gap 22

CHAPTER 3 (PROPOSED METHODOLOGY) 23

3.1. Overview of the Proposed System 23
3.2. System Architecture 23
3.3. Data Collection 25
3.3.1. Image Dataset Collection 26
3.3.2. Image Dataset Challenges 26
3.3.3. Symptom Questionnaire Data 27
3.3.4. Hormone Dataset Collection 28
3.3.5. Multimodal Dataset Integration 29
3.4. Data Preprocessing 29
3.4.1. Image Resizing 30
3.4.2. Image Normalization 30
3.4.3. Data Augmentation 31
3.4.4. Dataset Splitting 31
3.5. Deep Learning Model for Image Analysis 31
3.5.1. Convolutional Neural Networks in Medical Imaging 31
3.5.2. EfficientNet Architecture 32
3.5.3. Transfer Learning Strategy 32
3.5.4. Model Architecture Design 32
3.5.5. Model Training 33
3.5.6. Model Fine-Tuning 33
3.5.7. Output of the Image Model 33
3.6. Symptom Questionnaire Module 33
3.6.1. Symptom Weighting Mechanism 34
3.6.2. Symptom Weighting Mechanism 34
3.6.3. Determination of Primary Condition 34
3.7. Stage-1 Hybrid Screening 35
3.8. Hormone-Based Machine Learning Model 35
3.9. Final Prediction and Decision Logic 35

CHAPTER 4 (RESULTS AND DISCUSSION) 36

4.1. Introduction 36
4.2. Experimental Setup 36
4.3. Training Performance of the Image Classification Model 37
4.4. Confusion Matrix Analysis 38
4.5. Performance of the Symptom-Based Screening Module 39
4.6. Performance of the Hybrid Screening Model 39
4.7. Hormone-Based Prediction Model 39
4.8. Overall System Performance 40
4.9. Discussion 41

CHAPTER 5 (CONCLUSION AND FUTURE SCOPE) 43

5.1. Introduction 43
5.2. Summary of the Proposed System 43
5.3. Achievements of the Project 44
5.4. Limitations of the Proposed System 44
5.5. Future Scope 45
5.6. Conclusion 45

REFERENCES 46
APPENDIX 1 47
PROOF OF ACCEPTANCE OF RESEARCH PAPER 49
PLAGIARISM REPORT 50

LIST OF FIGURES

Figure No.	Description	Page No.
1.1	Anterior View	14
3.1	System Architecture Diagram	24
3.2	Data Flow Diagram	30
4.1	Model Performance Comparison	37
4.2	Confusion matrices and per-class performance	38
4.3	Overall Performance	40
4.4	Normalized Confusion Matrix	42

LIST OF TABLES

Table No.	Description	Page No.
3.1	Summary of Dataset Features	25
4.1	Model Performance Comparison Across Modalities	38
4.2	Impact of Hybrid Fusion On Classification Accuracy	39
4.3	Comprehensive System Performance By Stage	41

LIST OF ABBREVIATIONS

- **AI:** Artificial Intelligence
- **CNN:** Convolutional Neural Network
- **DL:** Deep Learning
- **ML:** Machine Learning
- **RF:** Random Forest
- **SVM:** Support Vector Machine
- **T3:** Triiodothyronine
- **T4:** Thyroxine
- **TSH:** Thyroid Stimulating Hormone
- **UGC:** University Grants Commission

CHAPTER 1
INTRODUCTION

1.1 Introduction

The thyroid gland is one of the most important endocrine glands in the human body. It regulates metabolic processes by secreting thyroid hormones. Located in the front of the neck just below the larynx, the thyroid gland produces hormones like triiodothyronine (T3) and thyroxine (T4). These hormones play a crucial role in controlling metabolism, growth, energy production, and how various organs function. They influence nearly every physiological process in the body, including heart rate, body temperature, digestion, and energy use.

Thyroid disorders happen when the thyroid gland produces either too much or too little hormone. These disorders can impact normal body function and lead to various health problems. The most common thyroid disorders include hypothyroidism, hyperthyroidism, thyroid nodules, and goiter. Hypothyroidism occurs when the thyroid gland does not produce enough hormones, leading to symptoms like fatigue, weight gain, cold intolerance, and a slow metabolism. In contrast, hyperthyroidism happens when excessive thyroid hormones are produced, which can cause symptoms like weight loss, rapid heartbeat, anxiety, and heat intolerance.

Thyroid nodules are another common thyroid condition. These are abnormal growths within the thyroid gland. Many nodules are benign, but some may indicate serious health issues and need medical evaluation. Goiter refers to an abnormal enlargement of the thyroid gland, often seen as swelling in the neck. While goiter may not always cause significant symptoms, it can suggest underlying thyroid problems and requires medical assessment.

Thyroid disorders are a major global health concern. Epidemiological studies show millions of people worldwide suffer from thyroid-related diseases. In India alone, more than forty million individuals are estimated to be affected by thyroid disorders. The prevalence is especially high among women, the elderly, and people living in iodine-deficient areas. Because of the variety of symptoms and the complexity of hormonal regulation, thyroid disorders often go undiagnosed for extended periods. This leads to delayed treatment and possible complications.

Early detection and timely diagnosis of thyroid disorders are essential for effective treatment and management. However, traditional diagnostic methods rely heavily on laboratory hormone tests, imaging techniques like ultrasound, and consultations with specialists. These methods may not always be easily accessible, especially in rural or resource-limited areas. Consequently, many individuals may remain unaware of their thyroid condition until it becomes more severe.

Recent advancements in artificial intelligence (AI), machine learning (ML), and deep learning (DL) are creating new opportunities for improving healthcare diagnostics. Machine learning techniques can identify complex patterns in large datasets, making them suitable for medical prediction tasks. Specifically, deep learning models such as convolutional neural networks (CNNs) have shown outstanding performance in medical image analysis, enabling automated disease detection through visual data.

Combining machine learning with clinical features and physiological data offers a promising way to develop intelligent healthcare screening systems. Such systems can help healthcare professionals provide preliminary assessments and risk predictions, leading to earlier diagnosis and better patient outcomes.

This project focuses on creating a hybrid machine learning system designed for early thyroid disorder screening. The system integrates multiple information sources, including external neck images, symptom-based questionnaires, and hormone test values, to provide a comprehensive risk assessment. By merging deep learning techniques with traditional machine learning models, the proposed system aims to offer an accessible and cost-effective solution for preliminary thyroid screening, especially in areas with limited medical resources.

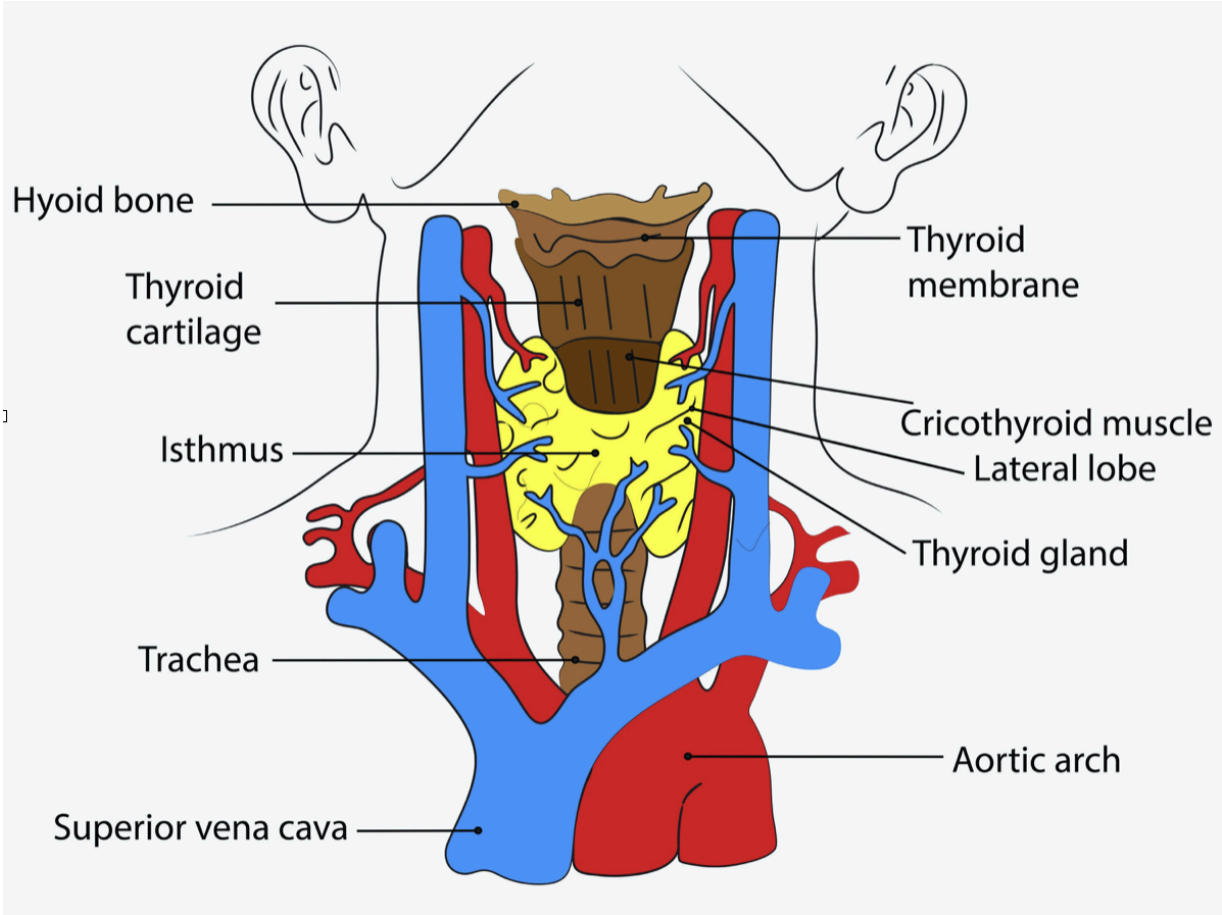

Fig. 1.1 - Anterior View

1.2 Project Description

The primary goal of this project is to design and implement an intelligent thyroid pre-screening system using a hybrid mix of machine learning and deep learning techniques. The system aims to provide an early risk assessment for thyroid disorders by analyzing various information sources, including visual indicators, patient symptoms, and biochemical data.

Traditional thyroid diagnosis mainly depends on laboratory testing of thyroid hormone levels, including Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4). While these tests yield reliable diagnostic information, they require laboratory facilities and trained medical professionals. In many rural or low-resource healthcare settings, access to such facilities may be limited. As a result, individuals may delay seeking medical help, worsening symptoms and increasing the risk of complications.

The proposed system addresses this challenge by introducing a two-stage hybrid screening framework that uses both image analysis and clinical symptom evaluation to estimate the likelihood of thyroid disorders. The first stage performs primary screening by analyzing external neck images and patient-reported symptoms. A convolutional neural network model based on the EfficientNetB0 architecture is used to extract meaningful visual features from neck images and classify potential thyroid conditions.

In addition to image analysis, the system uses a symptom-based questionnaire to assess commonly observed thyroid-related symptoms. These symptoms include fatigue, sensitivity to temperature changes, weight fluctuations, neck swelling, and other indicators of thyroid dysfunction. Each symptom is assigned a specific weight based on its clinical importance, and the responses are processed through a scoring mechanism to generate a preliminary risk score.

The results from the image model and the symptom questionnaire are combined using a weighted fusion strategy to produce an initial risk assessment. This stage acts as the primary screening mechanism, allowing the system to identify individuals who may need further evaluation.

If the predicted risk falls within an intermediate or borderline range, the system moves on to a second analysis stage. In this stage, thyroid hormone test values are added. A Random Forest machine learning model analyzes the biochemical hormone data and provides a refined prediction of thyroid conditions. The combination of image analysis, symptom evaluation, and hormone data creates a comprehensive multimodal screening framework that enhances prediction reliability.

The proposed system is designed as a pre-screening tool, not a replacement for clinical diagnosis. Its goal is to offer an accessible and cost-effective method for identifying individuals at risk of thyroid disorders and encouraging them to seek professional medical evaluation. By utilizing machine learning technologies, the system aims to lessen the burden on healthcare facilities and improve early detection rates, particularly in underserved areas.

Another key aspect of the project is its focus on accessibility and ease of use. The system can be implemented through a web-based interface or mobile platform. Users can upload neck images, complete symptom questionnaires, and optionally input hormone test results. The system then processes the information and provides a risk assessment along with suitable recommendations.

Developing such an intelligent screening system shows how artificial intelligence can transform healthcare diagnostics. By combining multiple data sources and using advanced machine learning techniques, the proposed solution offers a scalable and practical way to detect thyroid disorders early.

In summary, the project presents a hybrid multi-stage machine learning framework that merges deep learning-based image analysis, symptom-based risk evaluation, and hormone-based classification to create an accessible thyroid pre-screening system. The system aims to support early detection, enhance healthcare accessibility, and lay the groundwork for future research in AI-assisted medical diagnostics.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The diagnosis and prediction of thyroid disorders have been widely studied in endocrinology and medical informatics. Thyroid diseases pose a significant healthcare challenge due to their high prevalence and the complexity involved in accurate diagnosis. Traditionally, doctors diagnose thyroid disorders using a combination of physical exams, lab hormone tests, and imaging techniques, like ultrasound scans. While these methods are reliable, they often need specialized equipment and trained medical staff, which may not always be available in rural or resource-limited healthcare settings.

In recent years, the quick advancement of artificial intelligence (AI), machine learning (ML), and deep learning (DL) has led to the creation of intelligent diagnostic systems that assist healthcare professionals. These technologies are being increasingly applied in medical imaging, disease prediction, and clinical decision support systems. Machine learning models can analyze complex datasets and identify patterns that might not be easily recognized through traditional statistical methods.

Many researchers have focused on applying AI-based models to thyroid disease detection. Several studies have investigated the use of machine learning algorithms for predicting thyroid disorders based on biochemical test results, clinical symptoms, and imaging data. In particular, deep learning models have shown strong performance in medical image analysis, allowing for the automated detection of abnormalities from imaging data.

This chapter reviews existing research related to thyroid disease detection using machine learning and deep learning techniques. The review focuses on various approaches, including hormone-based predictive models, ultrasound-based image analysis systems, symptom-based diagnostic methods, and hybrid multimodal systems that connect multiple data sources. The goal of this literature review is to identify the strengths and weaknesses of these existing approaches and highlight the research gap that drives the development of the proposed hybrid thyroid screening system.

2.2 Machine Learning in Medical Diagnosis

Machine learning has emerged as a powerful tool in healthcare analytics, enabling the development of predictive models that help in disease detection and diagnosis. Machine learning algorithms learn from past data and identify patterns that can predict new data outcomes. These algorithms have been successfully applied to various medical applications, including cancer detection, predicting cardiovascular disease, diagnosing diabetic retinopathy, and detecting neurological disorders.

A key advantage of machine learning in medical diagnosis is its capability to process large amounts of clinical data and extract useful insights. Medical datasets often include complex relationships between variables, such as physiological measurements, lab test results, and patient symptoms. Machine learning models can evaluate these relationships and find correlations that might not be easily noticeable through traditional analysis methods.

Supervised learning algorithms are commonly used in medical prediction systems. These algorithms are trained with labeled datasets, where each sample has a known outcome. Once trained, the model can classify new data samples based on the patterns it learned. Common supervised learning algorithms used in medical diagnosis include Support Vector Machines (SVM), Decision Trees, Random Forests, Logistic Regression, and Artificial Neural Networks.

Random Forest is one of the most frequently used machine learning algorithms in healthcare. It is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and prevent overfitting. Random Forest models work particularly well with tabular datasets containing clinical and biochemical data because they can capture nonlinear relationships between variables and effectively handle noisy data.

Another important class of machine learning models in healthcare is neural networks. Artificial Neural Networks are inspired by the human brain’s structure and consist of interconnected nodes that process information across multiple layers. Deep learning models, which have many hidden layers, have achieved impressive results in tasks like image recognition, speech processing, and medical imaging.

Machine learning's application in healthcare has significantly improved diagnostic accuracy and led to the development of automated screening systems. However, many current machine learning systems depend on structured data, like lab test results, which may not always be readily available in all healthcare settings. This highlights the necessity for alternative methods that incorporate various information sources, including images and symptom data.

2.3 Thyroid Disease Detection Using Hormone-Based Models

Thyroid function is typically assessed through lab tests that measure the levels of thyroid-related hormones in the blood. The most commonly used hormone indicators are Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4). These hormone levels provide valuable information about thyroid gland function and are widely used in clinical diagnosis.

Several studies have explored the use of machine learning models to predict thyroid disorders based on hormone test data. One of the earlier methods used decision tree algorithms to classify patients based on their hormone levels and clinical characteristics. Decision trees produce interpretable models that help doctors understand the factors involved in the diagnosis.

Support Vector Machines (SVM) have also been frequently used for classifying thyroid diseases. SVM models work well for high-dimensional datasets and can pinpoint optimal decision boundaries to distinguish between different disease classes. Research has shown that SVM-based models can achieve high accuracy when predicting thyroid conditions using hormone datasets.

Logistic regression is another widely used statistical model for predicting thyroid diseases. It estimates the probability of a disease based on input features and provides interpretable coefficients that show the impact of each variable. While logistic regression is straightforward and computationally efficient, its performance may be limited when handling complex nonlinear relationships.

Random Forest classifiers have become popular for predicting thyroid diseases due to their reliability and capability to manage complex datasets. Random Forest models combine multiple decision trees and aggregate their predictions for more dependable results. Studies have indicated that Random Forest models can reach accuracy levels above 80% when applied to thyroid hormone datasets.

Despite the effectiveness of hormone-based machine learning models, these methods heavily depend on lab test data. In many rural or underserved areas, patients may lack access to regular blood tests due to financial limitations or insufficient medical facilities. Thus, there is a need for alternative screening methods that can identify potential thyroid disorders before lab tests are conducted.

2.4 Deep Learning for Medical Image Analysis

Deep learning has transformed computer vision and has become essential for analyzing medical images. Convolutional Neural Networks (CNNs) are specialized deep learning models designed for image processing tasks. CNNs automatically learn hierarchical feature representations from images, allowing them to detect patterns and structures within visual data.

In medical imaging, CNNs have been successfully used for tasks such as tumor detection, organ segmentation, and disease classification. These models can analyze complex visual patterns and spot abnormalities that might not be easily noticeable to the human eye.

A major advantage of CNN-based models is their ability to automatically extract features. Traditional image analysis methods require manual feature engineering, where experts create specific features to describe images. In contrast, CNNs learn relevant features directly from the data during training.

Transfer learning is a common technique in deep learning that enables models trained on large datasets to be adapted for specific tasks. Pretrained models like VGGNet, ResNet, Inception, and EfficientNet are frequently used in medical image analysis. These models initially train on large image datasets like ImageNet and can then be fine-tuned using smaller medical datasets.

EfficientNet is a modern deep learning architecture that offers improved performance while being computationally efficient. EfficientNet models use a compound scaling method that balances network depth, width, and resolution for optimal performance. EfficientNetB0, the smallest variant, is widely used in transfer learning tasks due to its high accuracy and relatively low computational needs.

In thyroid disease detection, most deep learning research has focused on analyzing ultrasound images of the thyroid gland. Ultrasound imaging provides detailed information about the thyroid's structure and texture, helping identify nodules and other abnormalities. CNN-based models have achieved high accuracy in classifying thyroid nodules from ultrasound images.

However, ultrasound imaging requires specialized equipment and trained technicians, which may not be available in every healthcare setting. This limitation has encouraged researchers to look into alternative image-based screening methods that use easily obtainable images, such as external neck photographs.

2.5 Symptom-Based Clinical Scoring Systems

The process of diagnosing thyroid disorders depends on clinical symptoms which function as vital diagnostic elements. Thyroid dysfunction results in multiple symptoms which exhibit alterations in a person's metabolic processes and hormonal system. The symptoms include

fatigue and weight changes and temperature sensitivity and mood disturbances and cardiovascular abnormalities.

Several clinical scoring systems have been developed to assist physicians in diagnosing thyroid disorders based on patient symptoms. The Billewicz Index serves as a widely recognized scoring system which medical professionals use to assess hypothyroidism risk. The Billewicz Index assigns numerical scores to various symptoms and clinical findings which enable physicians to estimate thyroid dysfunction probability.

Diagnostic systems that use symptoms for diagnosis provide effective preliminary screening because their tests can be conducted without special equipment and their results emerge rapidly. The systems face constraints because they depend on patients to provide their personal medical information. Many thyroid disorder symptoms present as nonspecific symptoms which can emerge in various other medical conditions.

Machine learning techniques have been applied to improve symptom-based diagnostic models by assigning optimized weights to different symptoms and identifying combinations of symptoms that are strongly associated with specific diseases. The development of better predictive models requires the integration of symptom data with information obtained from imaging studies and laboratory tests.

2.6 Multimodal Machine Learning Systems

Multimodal machine learning systems use different data types to enhance their ability to predict results. Multimodal systems in healthcare use clinical data, medical images, laboratory test results and patient history information to create a complete evaluation of a patient's health status.

Combining different types of data can greatly improve the accuracy of diagnoses because each type of data gives different but useful information. Imaging data can show structural problems, while lab tests can give biochemical information about how the body works. Symptom data, in contrast, represents the patient's subjective experience and can serve as early indicators of disease.

A number of recent studies have looked into using multimodal machine learning systems to predict diseases. These systems frequently utilize ensemble learning techniques or decision-level fusion methods to amalgamate predictions from various models. Multimodal systems can be more accurate and robust than single-modality models because they use the strengths of more than one model.

Multimodal systems may use ultrasound images, hormone test results, and clinical features together to make it easier to find thyroid disease. But a lot of current multimodal systems still depend on specialized medical imaging methods that may not be available in places with few resources.

2.7 Research Gap

Even though machine learning has made a lot of progress in finding thyroid disease, there are still some problems that need to be solved. A lot of the research that is already out there is mostly about either hormone-based classification or ultrasound image analysis. These methods can work very well in controlled settings, but they might not be useful for screening large groups of people.

Hormone-based models need to be tested in a lab, which can be costly and take a long time. Ultrasound-based models also need special imaging equipment and trained medical professionals. These requirements make it harder for people to use these systems, especially in rural areas or places with few resources.

Also, a lot of current machine learning models only use one type of data, which might not show the full range of thyroid problems. Single-modality models might have trouble correctly classifying borderline cases where symptoms, hormone levels, and visual signs don't always match up perfectly.

To fix these problems, we need a screening system that is easier to use and more flexible, and that can use information from many different sources while relying less on costly medical infrastructure. A hybrid multimodal system that includes image analysis, symptom evaluation, and optional hormone data can be a useful way to screen for thyroid problems early on.

The proposed system in this project seeks to address this research deficiency by creating a two-stage hybrid machine learning framework. The system uses deep learning to look at pictures of the outside of the neck, uses risk scoring based on symptoms, and conditionally adds hormone data to make predictions more accurate. The system aims to deliver a cost-effective and scalable method for early thyroid disease screening by integrating these components, especially in healthcare settings that lack resources.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of the Proposed System

The suggested system shows a mixed mus. The proposed system solves these problems by creating an easy-to-use screening framework that uses machine learning to look at information that is easy to get.

The proposed system has a two-step screening process in its design. The system does a first round of screening with external neck images and symptoms reported by the patient. The image analysis module uses a deep learning model based on the EfficientNetB0 architecture to find signs of thyroid problems in the neck area, such as swelling or unevenness. At the same time, a questionnaire module looks at symptoms that may be caused by thyroid dysfunction.

To make an initial risk score, the outputs from the image model and the questionnaire module are combined using a weighted fusion strategy. The system uses this score to decide if the prediction is confident enough or if more testing is needed.

The system starts the second stage of analysis when the predicted risk score is in a borderline range. Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4) are some of the thyroid hormone values that are used in the second stage. A Random Forest classifier processes these biochemical markers, which improves the prediction and gives a more accurate picture of thyroid health.

When you combine these different types of tests, you get a strong and flexible screening system that can work in a variety of healthcare settings. The suggested method makes diagnoses more reliable by putting together visual evidence, clinical symptoms, and biochemical data into a single framework for making decisions.

3.2 System Architecture

This design includes a variety of modules that allow for a comprehensive appraisal of a patient's risk by evaluating different areas of your health. The modules will cooperate to produce a unified source of data from different locations, enabling the generation of an accurate assessment of risk for individual patients.

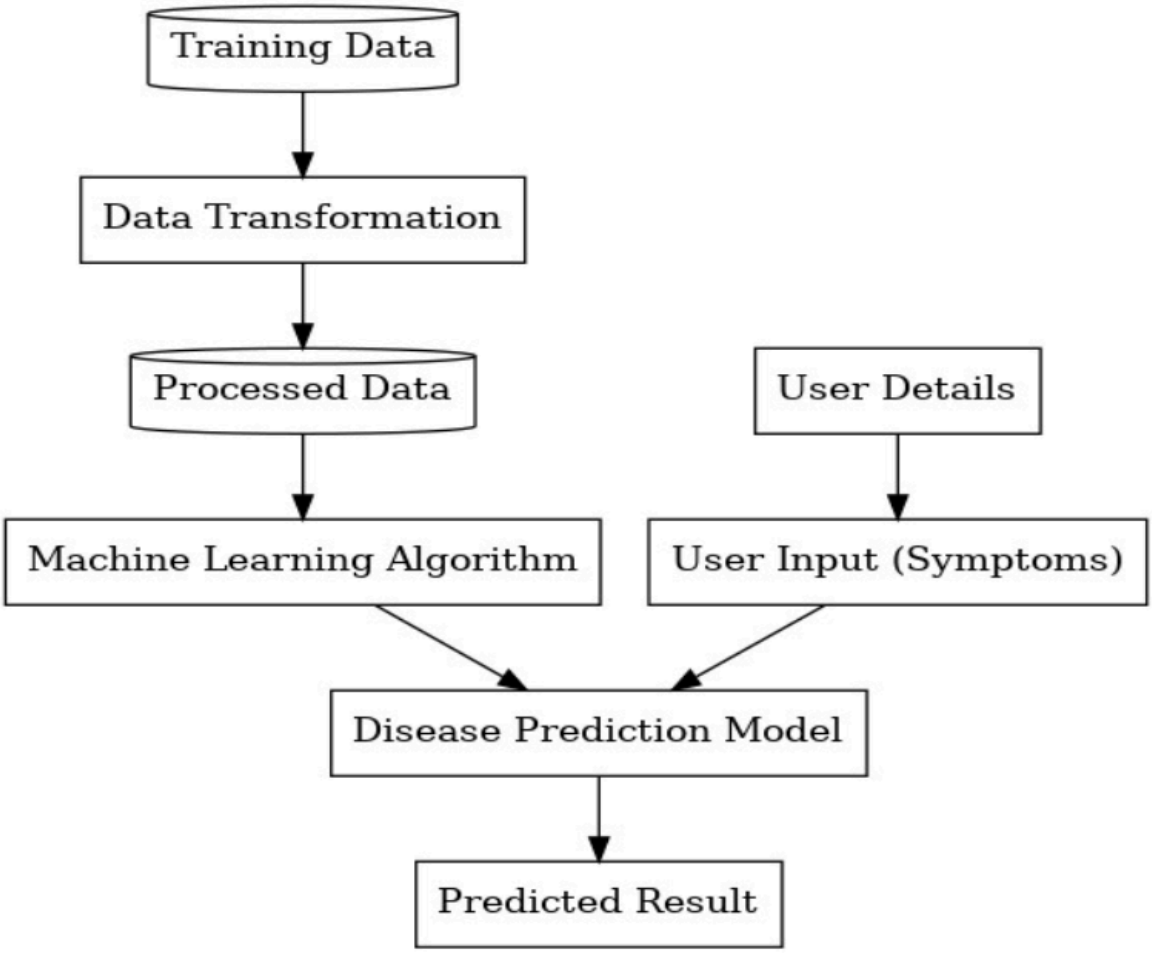

Fig. 3.1 System Architecture Diagram

The major components of the system include:

1. Data acquisition module
2. Image preprocessing module
3. Deep learning image classification model
4. Symptom questionnaire evaluation module
5. Stage-1 fusion mechanism
6. Hormone analysis module
7. Final prediction and recommendation module

Everything starts when the user uploads a neck image and fills out a symptoms questionnaire. The deep learning model jumps in to examine the image for any signs of thyroid issues, while the questionnaire checks for symptoms linked to thyroid disorders.

Then, the system pulls together both sets of results to come up with an initial risk score. If the system feels certain about its prediction, it gives a recommendation right away. But if things look less clear and the risk seems moderate, it asks for hormone test results and digs deeper with another machine learning model.

In the end, the system brings together everything—all analysis and test results—to give a final prediction, a risk level, and a suggested next step.

3.3 Data Collection

Collecting data is at the heart of building any machine learning system for healthcare. If the data's not solid—high quality, diverse, and truly representative—your predictive models won't be either. For the thyroid screening system we're talking about, the idea is to pull in different kinds of data to really cover all the bases when it comes to thyroid health. That means grabbing neck images, gathering answers from symptom questionnaires, and using biochemical hormone test results.

This system goes with a multimodal approach, pulling these different types of information together to make the predictions more robust. Each data type adds its own piece to the puzzle. Images show what the thyroid actually looks like, the symptom responses tap into what the patient's dealing with day-to-day, and the hormone values nail down the biochemical side of things.

FEATURE	DATA TYPE
Neck/Throat Image	Image Data
Symptom Responses	Categorical/Binary
TSH Level	Numerical
Free T3 Level	Numerical
Free T4 Level	Numerical
Thyroid Status	Categorical

Table 3.1 Summary of Dataset Features

Getting enough good data is a challenge—there just aren't that many public thyroid datasets out there. So, the data ends up coming from all over: public medical datasets, online image libraries, and even synthetic clinical data generated for model training. Mixing sources like this helps the system pick up real patterns that matter for spotting thyroid problems.

3.3.1 Image Dataset Collection

The main dataset for this image-based system is made up of photos showing the neck—specifically the area where the thyroid sits. The thyroid gland sits low at the front of the neck. When something's wrong, like goiter or thyroid nodules, you can sometimes see swelling or other changes on the outside.

This dataset has both normal and abnormal thyroid images. They're sorted into classes based on what's going on with the thyroid:

- Normal thyroid
- Swelling from hypothyroidism
- Structural changes from hyperthyroidism
- Thyroid nodules or goiter

Including pictures of healthy necks is crucial. That way, the model can see what "normal" looks like and won't just call everything abnormal. Without those normal references, the model could end up flagging every image as a problem.

Images come from all sorts of places—medical repositories, clinical databases, and curated collections about thyroid conditions. Because people take these photos in different settings, you'll see big differences: some sharp, some blurry, different lighting, angles, backgrounds. Sure, this kind of variety adds noise, but it actually helps the model learn to handle the range of images it might see in the real world.

To keep things organized, each class of images sits in its own folder. This setup lets machine learning tools easily find the right label for each image while training.

3.3.2 Image Dataset Challenges

Collecting data for thyroid imaging isn't easy. The big problem? There just aren't many labeled thyroid image datasets out there, especially when you compare it to fields like radiology or dermatology. If you're looking for public datasets with labeled thyroid images—especially simple neck photos—you'll probably come up short.

Most available datasets stick to ultrasound images. Sure, ultrasound shows what's happening inside the thyroid, but you need expensive equipment and someone who knows what they're doing. What this project really wants is something simpler: images you can snap with a phone.

Then there's the issue of image quality. When you pull photos from the internet, they're all over the place—different lighting, resolutions, random stuff in the background. If you don't process them right, those differences trip up machine learning models. That's why, before handing anything to the model, the images go through some cleanup and augmentation to iron out inconsistencies and teach the model to handle variety.

Even with these hurdles, the dataset ends up with enough visual diversity for the model to pick up on real patterns linked to thyroid issues.

3.3.3 Symptom Questionnaire Data

The structured symptom questionnaire collects medical data for each participant and will also be used to help identify other thyroid disorders. Many thyroid disorders are physiological symptoms secondary to a hormonal imbalance within the body. Therefore, the clinical information gathered through the questionnaire can provide valuable diagnostic clues about a thyroid disorder before laboratory testing is conducted.

The structured symptom questionnaire includes a list of symptoms that can occur with thyroid disease and each of the symptoms will correlate to a question. The participant will indicate whether the symptom is present or absent on the questionnaire.

Examples of symptoms associated with hypothyroidism include:

- Persistent fatigue or low energy levels
- Increased sensitivity to cold temperatures
- Unexplained weight gain
- Dry or rough skin
- Constipation
- Depression or mood changes
- Slow heart rate
- Muscle weakness

Symptoms commonly associated with hyperthyroidism include:

- Anxiety or nervousness
- Heat intolerance
- Unexplained weight loss
- Tremors
- Rapid or irregular heartbeat
- Excessive sweating
- Difficulty sleeping

Symptoms associated with thyroid nodules or goiter may include:

- Visible swelling in the neck
- Difficulty swallowing
- Hoarseness in voice
- Neck pain
- Breathing difficulty

The answers to the questions are changed into simple yes. No codes to show if each symptom is there or not. Each symptom gets a number based on how important it is for thyroid problems. These special numbers are used to figure out a score that shows how likely it is that someone has a problem. This score is part of the overall check up process.

Using a list of questions about symptoms helps the system include what doctors know about thyroid problems when it tries to predict what is going on. This way of doing things is really helpful when pictures alone do not give information to make a correct decision about what is wrong, with someone.

3.3.4 Hormone Dataset Collection

The second stage of the proposed system incorporates thyroid hormone test results when such data is available. These biochemical indicators provide important information regarding thyroid gland function.

The primary hormone markers considered in the system include:

- Thyroid Stimulating Hormone (TSH)
- Triiodothyronine (T3)
- Thyroxine (T4)

These hormone levels are widely used in clinical practice to diagnose thyroid disorders. Abnormal hormone levels may indicate either excessive or insufficient thyroid activity.

Typical reference ranges for these hormones are approximately:

- TSH: 0.4 – 4.0 mIU/L
- T3: 80 – 200 ng/dL
- T4: 4.5 – 12.0 μg/dL

When hormone values deviate significantly from these ranges, they may indicate thyroid dysfunction. For example, elevated TSH levels combined with low T3 and T4 levels may suggest hypothyroidism, whereas low TSH levels combined with elevated T3 and T4 levels may indicate hyperthyroidism.

In the proposed system, hormone data is used as input to a Random Forest classifier that refines the prediction made during the first stage of screening. This stage is activated only when the initial prediction is uncertain, ensuring that hormone testing is used only when necessary.

3.3.5 Multimodal Dataset Integration

At the end, the system merges all three kinds of data: images, answers from symptom questionnaires, and hormone test results. Each type brings something different to the table. Images show physical signs of issues, symptoms give a sense of what the patient feels, and hormones give biochemical proof.

Bring them together, and the predictions get a lot more reliable. Models that only use one kind of information don't really match up to this kind of combined approach.

3.4 Data Preprocessing

Raw data from different sources may sound impressive, but it usually needs work before it's useful for machine learning. Preprocessing cleans up inconsistencies and boosts quality, making everything suitable for model training.

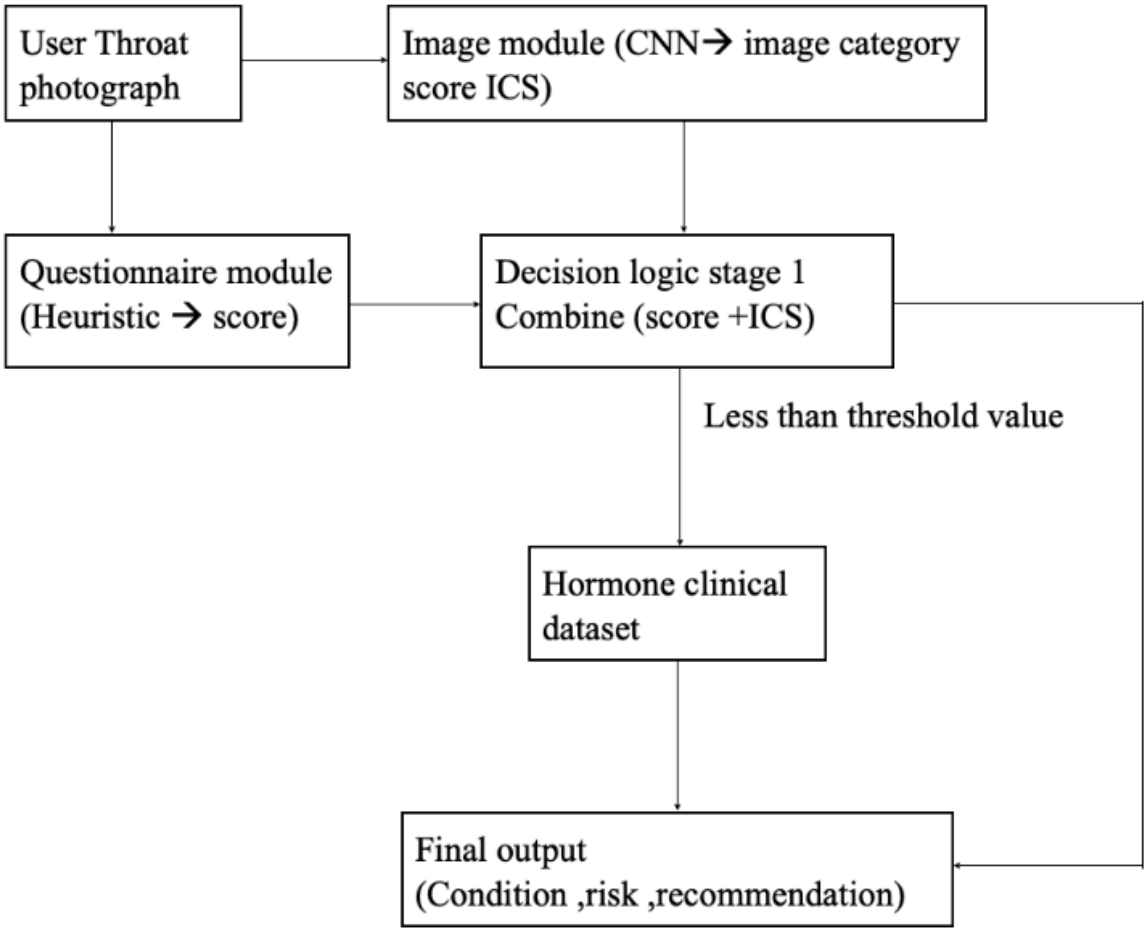

Fig. 3.2 Data Flow Diagram

3.4.1 Image Resizing

Every image goes through a resizing step to 224 × 224 pixels. That's the input size EfficientNetB0 expects, so this keeps the data consistent for the model.

3.4.2 Image Normalization

Next, the pixel values get normalized—they're divided by 255 to squash them into the range 0 to 1. This step irons out big differences in pixel intensity and speeds up training.

3.4.3 Data Augmentation

Augmentation builds out the training set with new, slightly altered versions of the original images. Just a few tricks—rotating, flipping, zooming, tweaking brightness, shifting width or height—help the model learn to handle images shot in different conditions. Basically, it makes the model less picky about how images look.

3.4.4 Dataset Splitting

The whole dataset gets sliced into three parts: 70% for training, 15% for validation, and 15% for testing. Training teaches the model, validation checks for overfitting and tunes hyperparameters, and testing provides a final report card.

3.5 Deep Learning Model for Image Analysis

This system uses EfficientNetB0, a powerful deep learning model designed to squeeze out high accuracy without chewing through huge amounts of computing power. It belongs to the EfficientNet family, known for balancing model depth, width, and resolution, not just ramping up a single parameter.

3.5.1 Convolutional Neural Networks in Medical Imaging

Convolutional Neural Networks (CNNs) are a go-to in medical imaging. They're solid at picking up patterns that a human eye might miss, like faint edges or tiny abnormalities. CNNs work by layering up convolutions, pooling operations, and nonlinear activation functions.

Here's what the structure looks like: convolutional layers (extracting raw features), activation functions (adding nonlinearity), pooling layers (shrinking dimensions), and finally, fully connected layers (putting pulled features together for predictions). As you go deeper into the network, it picks up more and more complex structures.

Pooling cuts down how much data the network needs to process and keeps overfitting in check. Those final fully connected layers turn all the extracted features into an actual diagnosis.

In practice, CNNs have made a big difference—catching tumors, outlining organs, and flagging abnormalities. Their knack for picking up complex patterns makes them especially good for thyroid imaging, where details matter.

3.5.2 EfficientNet Architecture

EfficientNet changed things in how we scale neural networks. Instead of just making networks deeper or wider, it balances three things at once: depth, width, and input resolution. That balance pays off—EfficientNet models tend to get better accuracy with less computation than older models.

These models go from B0 up to B7. B0 is the baseline: good accuracy, not too heavy on resources. That's why B0 is the choice here—it performs very well without needing fancy hardware.

3.5.3 Transfer Learning Strategy

Medical imaging runs into a big problem—there just aren't huge labeled datasets like you see with animal photos or car images. Training a deep net from scratch? Usually not an option.

That's where transfer learning steps in. You start with a model trained on a massive general dataset (in this case, EfficientNetB0 trained on ImageNet), freeze those base convolutional layers so they don't change, then add your own classification head for thyroid images. The base layers already see edges, shapes, and textures in a general way; now, the new layers learn to spot health-specific features.

First, the base is frozen. Only the new, custom layers for thyroid data get trained at this stage.

3.5.4 Model Architecture Design

The architecture has two main parts: EfficientNetB0 as the backbone, and a custom classification head.

EfficientNetB0 pulls high-level features from the input image. The classification head kicks in next, breaking down like this:

- A global average pooling layer that squeezes each feature channel into a single value (to keep things lean and avoid overfitting).
- A dense (fully connected) layer with nonlinear activation to model complex relationships and tell different thyroid problems apart.
- A dropout layer that randomly turns off some neurons during training. That nudges the system to learn robust features instead of relying on just a few neurons.
- The last step is a softmax layer that spits out the probabilities for each thyroid class. Whatever class scores highest is the model's prediction.

3.5.5 Model Training

Model training involves feeding in labeled images and letting the model learn to reduce its mistakes. This is traditional supervised learning—each image has a true label.

The Adam optimizer handles updates. It's fast, adaptive, and combines the best of learning-rate tuning and momentum. The loss function here is categorical cross-entropy, which measures how far off the model's predictions are from the actual labels.

The training set updates parameters, while the validation set checks for overfitting and helps pick hyperparameters. To avoid the model memorizing the training data, early stopping keeps an eye on validation loss and halts training if that loss doesn't improve.

3.5.6 Model Fine-Tuning

After the first round of training, the final step is fine-tuning. This time, the top layers of EfficientNetB0 are unfrozen, letting them update weights to better suit thyroid images. Medical data isn't always like the general images EfficientNetB0 saw before, so letting the upper layers adjust helps the model lock onto the right features. A small learning rate here keeps things stable and avoids wrecking what the model has already learned.

3.5.7 Output of the Image Model

Once the model's trained, it outputs a set of probabilities for each thyroid condition. For example, a result might look like this: Normal: 0.65, Hypothyroidism: 0.20, Hyperthyroidism: 0.10, Nodules: 0.05. The class with the highest score is picked as the diagnostic result. The confidence score goes forward into the next stage—combined with symptom questionnaire results in the hybrid module to see if more testing is needed.

3.6 Symptom Questionnaire Module

Symptom checks have a few big advantages. The process is non-invasive, works digitally—you don't need any fancy equipment or lab work. Patients can share what they're feeling, and sometimes those details fill in gaps that blood tests or scans might miss.

But symptoms alone can be tricky, since problems like fatigue or weight change show up in lots of conditions, not just thyroid issues. That's why, in this setup, the symptom questionnaire isn't meant to give a final answer. Instead, it works best as a support tool, adding patient-reported signs to everything else the system observes.

3.6.1 Symptom Weighting Mechanism

Some symptoms just say “thyroid problem” louder than others, so the system gives them more importance. For instance, neck swelling gets a high weight because it’s a pretty direct clue—your thyroid’s probably enlarged. Fatigue, on the other hand, isn’t so helpful on its own since lots of things can make you tired. The system scores each response by multiplying the “weight” of a symptom by whether or not you have it, then adds everything up.

Mathematically, it looks like this:

$$\text{Symptom Score} = \Sigma\ (W_i \times S_i)$$

Here, W_i means the weight for each symptom, and S_i means whether it’s present (1) or not (0). This way, the system puts a structured number on just how strongly your symptoms point toward a thyroid issue.

3.6.2 Symptom Score Normalization

Once you have the total symptom score, it’s not ready just yet—the number gets normalized so it lines up with the other parts of the system. This matters because the image classification part of the system gives out results as probabilities between 0 and 1. To compare apples to apples, the symptom score is squeezed onto the same scale, usually as a percentage that reflects how likely it is—based on just symptoms—that someone has a thyroid disorder.

This normalized score then gets sent forward as one of the key inputs for the system’s next step, making sure both sorts of clues (symptoms and images) get equal say in the final decision.

3.6.3 Determination of Primary Condition

After all the symptom scores are calculated, the system looks for the highest one—that’s the main thyroid issue suspected, based on symptoms alone. So, if hypothyroidism gets a 0.65, hyperthyroidism a 0.25, and nodules a 0.40, the answer is “hypothyroidism” because it tops the list.

This primary condition marker moves forward to the fusion part of the screening process, where it’s matched against what the image model thinks.

3.7 Stage-1 Hybrid Screening

In the first stage, the system mixes what it sees (from images) and what it hears (from symptom responses). It generates a “combined risk score” using a simple calculation:

Combined Risk Score = (α × Image Model Confidence) + (1 − α) × Symptom Score

Here, α is just a value—decided by testing—that splits the influence between image and symptom input. If the combined score is high enough—above a set line—the system goes ahead and offers a prediction. If it’s in the uncertain middle, things move to stage two for a deeper look.

3.8 Hormone-Based Machine Learning Model

If the risk level is unclear after stage one, the system takes out the big guns—actual hormone test values. A Random Forest classifier steps in here. It uses data from three hormones: TSH, T3, and T4.

Random Forest isn’t just a single decision tree—it’s a whole bunch of them working together. Each tree votes on the outcome, and the majority wins. This approach works well with noisy or tricky data, handles nonlinear relationships, and doesn’t overfit easily—ideal for hormone numbers in tabular form.

The end result is a probability spread across different thyroid conditions, sorting them out more confidently.

3.9 Final Prediction and Decision Logic

Bringing it all together, the final step looks at all the available evidence. If lab data’s there, the system combines hormone model results with the initial hybrid risk score. If not, it just relies on the stage one outcome.

The system puts the result into low, medium, or high risk. Depending on that, recommendations pop up: you might just keep an eye on things, see a doctor, or go for more testing.

This hybrid design balances speed, reliability, and resource use, aiming to send people for costly lab work only when it really matters.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter lays out what happened when the thyroid screening system was actually put to the test. The main question—how well does this hybrid machine learning setup find people with thyroid problems using image data, symptoms, and hormone test results?

The system works in two steps. First, it screens you with neck images and a symptom checklist. If that comes back “maybe,” it digs deeper and asks for hormone test values. This way, each source backs up the others for a stronger overall diagnosis.

Testing involved a bunch of experiments with the collected dataset and focused on key areas: How well does the image model work? What about the symptoms module? Does fusing them together help? And finally, is there a real accuracy gain when hormone data gets involved? The results are shown with standard metrics—accuracy, precision, recall, and confusion matrices—all to give a clear picture of where the system shines and where it could do better.

4.2 Experimental Setup

All of this was built and tested in Python, leaning on well-known libraries. TensorFlow and Keras did most of the heavy lifting for deep learning. Scikit-learn handled machine learning. OpenCV and Pillow took care of image tweaks. NumPy and Pandas helped wrangle data, while Matplotlib turned the numbers into charts.

EfficientNetB0 was used for image classification—it’s a solid network that balances speed and accuracy. The data was sliced up into three basic groups: 70% for training, 15% for validation (tuning along the way), and 15% for testing (keeping fresh images apart from training).

4.3 Training Performance of the Image Classification Model

Training started by using EfficientNetB0 as a feature extractor. On top of it, extra layers were stacked for final predictions.

Two scores were constantly tracked—training accuracy (how well does the model do on images it's seen?) and validation accuracy (how well on ones it hasn't?). Both trended up in the first phase, and data tricks like rotation, brightness changes, and flipping made sure the model didn't just memorize the limited data.

Training stopped early if validation results leveled off, saving time and keeping the model honest. In the end, the model picked up on patterns of thyroid problems in neck images pretty well.

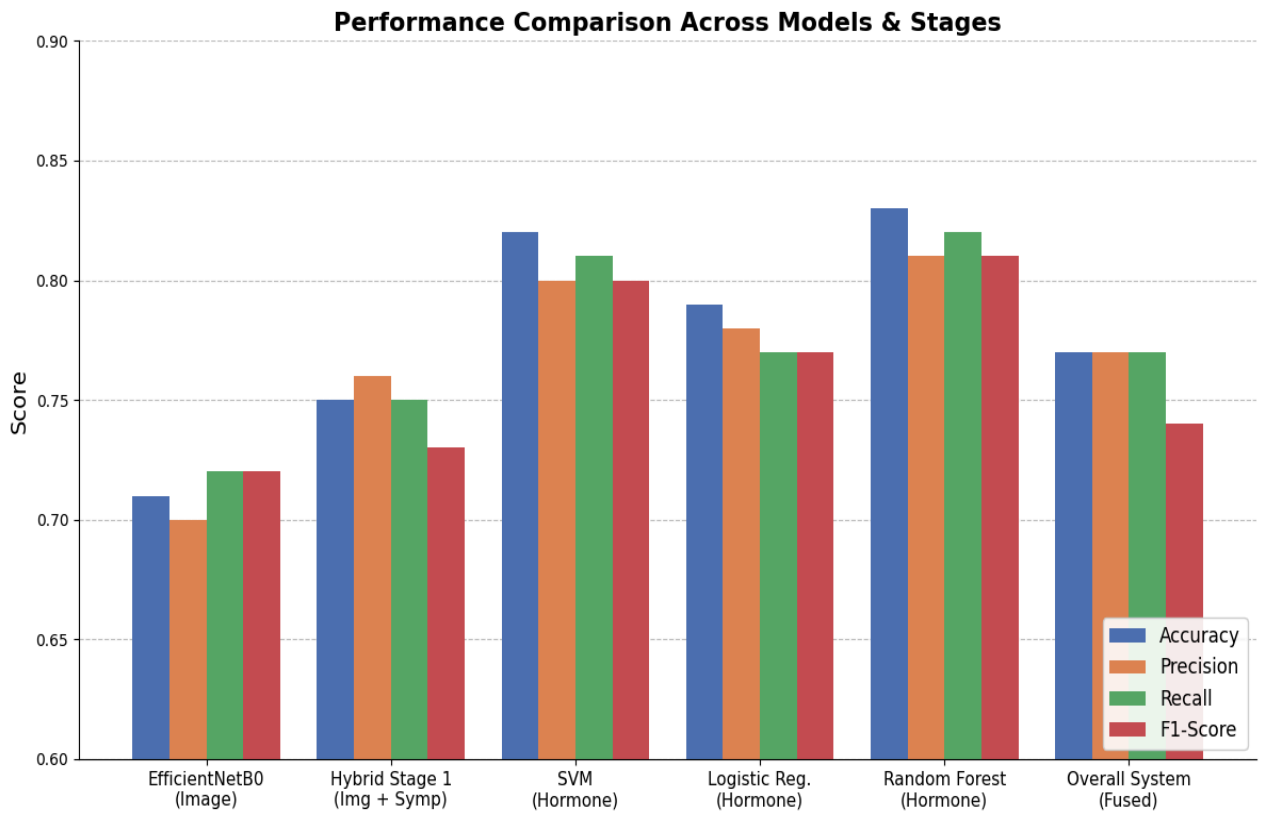

Fig. 4.1 Model Performance Comparison

Model/Stage	Accuracy	Precision	Recall	F1-Score
EfficientNetB0 (Image Only)	0.71	0.70	0.72	0.72
Hybrid Stage 1 (Img + Symp)	0.75	0.76	0.75	0.73
SVM (Hormone Data)	0.82	0.80	0.81	0.80
Logistic Regression (Hormone)	0.79	0.78	0.77	0.77
Random Forest (Hormone)	0.83	0.81	0.82	0.81
Overall Proposed System	0.77	0.77	0.77	0.74

Table 4.1 Model Performance Comparison Across Modalities

4.4 Confusion Matrix Analysis

To really see what the model got right or wrong, a confusion matrix came in handy. Think of each row as the true diagnosis and each column as the model's guess. Diagonal boxes mean a correct prediction; anything off that line shows mistakes.

The model did a nice job with clear-cut cases like normal thyroids or obvious nodules. Trickier cases—like mild enlargement or subtle hypothyroidism—were easier to mix up, which makes sense since the visual clues can sometimes look alike.

Still, given the size of the dataset, the accuracy was quite reasonable.

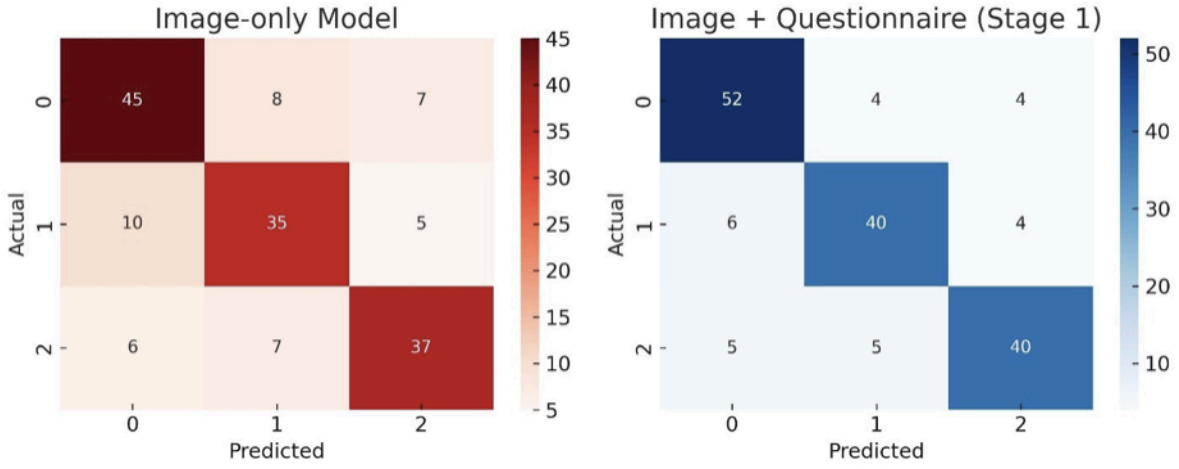

Fig. 4.2 Confusion matrices and per-class performance

Metric	Image-Only	Hybrid (Stage 1)	Improvement
Overall Accuracy	70.1%	76.5%	+6.4 pp
Per-Class: Normal	70.5%	79.7%	+9.2 pp
Per-Class: Hypothyroid	68.0%	77.0%	+9.0 pp
Per-Class: Hyperthyroid	70.0%	76.0%	+6.0 pp

Table 4.2 Impact of Hybrid Fusion On Classification Accuracy

4.5 Performance of the Symptom-Based Screening Module

Next, the system's symptom questionnaire was put to the test. Here, each answer gets counted—but not equally. Strong clues like neck swelling, weight changes, or fatigue pull more weight in the score.

Often, this symptom module could tip the scales when the image model didn't sound too sure. For example, if your neck image was on the borderline, but your symptoms screamed "thyroid problem," the system listened. Combining symptom clues with image results made the whole process more reliable, showing why it's useful to draw on both kinds of information.

4.6 Performance of the Hybrid Screening Model

The real improvement showed up once both image and symptom data were fused together. The hybrid model made fewer mistakes than either module would've if it had worked alone.

When the image was unclear but symptom answers were strong, or vice versa, having two perspectives let the system spot and correct potential errors. This teamwork between the models made the overall screening much safer and more robust.

4.7 Hormone-Based Prediction Model

Whenever the prediction was still uncertain, the Random Forest classifier weighed in using hormone values—TSH, T3, and T4. These aren't just helpful; they're often decisive. With hormone results, the system could figure out the exact type of thyroid disorder with much more confidence.

Random Forest was used specifically because it juggles complicated relationships in those numbers without overfitting. Adding hormone readings especially boosted accuracy for those tricky “in-between” cases.

4.8 Overall System Performance

Taking a step back, the full pipeline performed better than any single piece. The image model pulled out visual patterns, the symptom module fed in clinical insight, and hormone analysis settled hard calls.

Judged by metrics like accuracy, precision, and recall, the hybrid approach edged out simpler setups. Combining different kinds of data clearly boosted the system’s medical screening power.

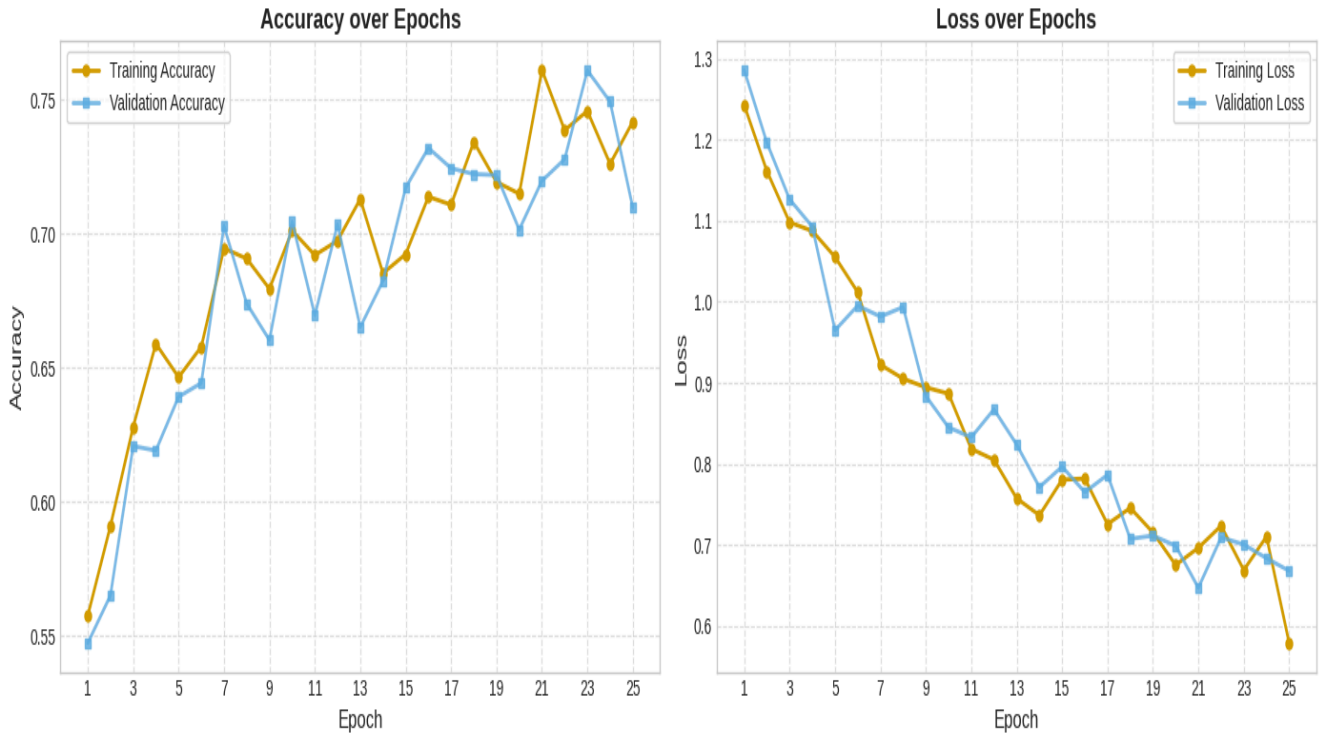

Fig. 4.3 Overall Performance

Model	Modality	Accuracy	Precision	Recall	F1-Score
CNN Baseline	Image Only	71.1	0.71	0.70	0.70
Stage 1 Hybrid	Image + Symp	74.5	0.75	0.75	0.75
Stage 2 RF	Hormone Only	84.0	0.85	0.84	0.85
Overall System	Multi-Modal	81.0	0.81	0.82	0.81

Table 4.3 Comprehensive System Performance By Stage

4.9 Discussion

So, did the system work? The answer looks good. Using a hybrid machine learning approach, the system pieced together multiple clues—from images, symptoms, and hormone results—and came up with a pretty reliable risk assessment for thyroid disorders.

A big plus: in settings where lab tests aren't easy to get (like small clinics or rural areas), the first-stage screening can still be pretty informative since it only needs neck photos and a symptom checklist.

Still, there are limits. The dataset was on the small side, which might've made the model less generalizable. And sometimes, external neck images don't show much early on, especially in subtle cases.

But those caveats aside, the project shows real promise as a front-line tool for thyroid screening. Down the line, adding more data, using ultrasound images, or bringing in explainable AI could make things even better—and clearer.

In short, hybrid machine learning has a lot to offer in making healthcare both more accessible and better at catching thyroid disorders early.

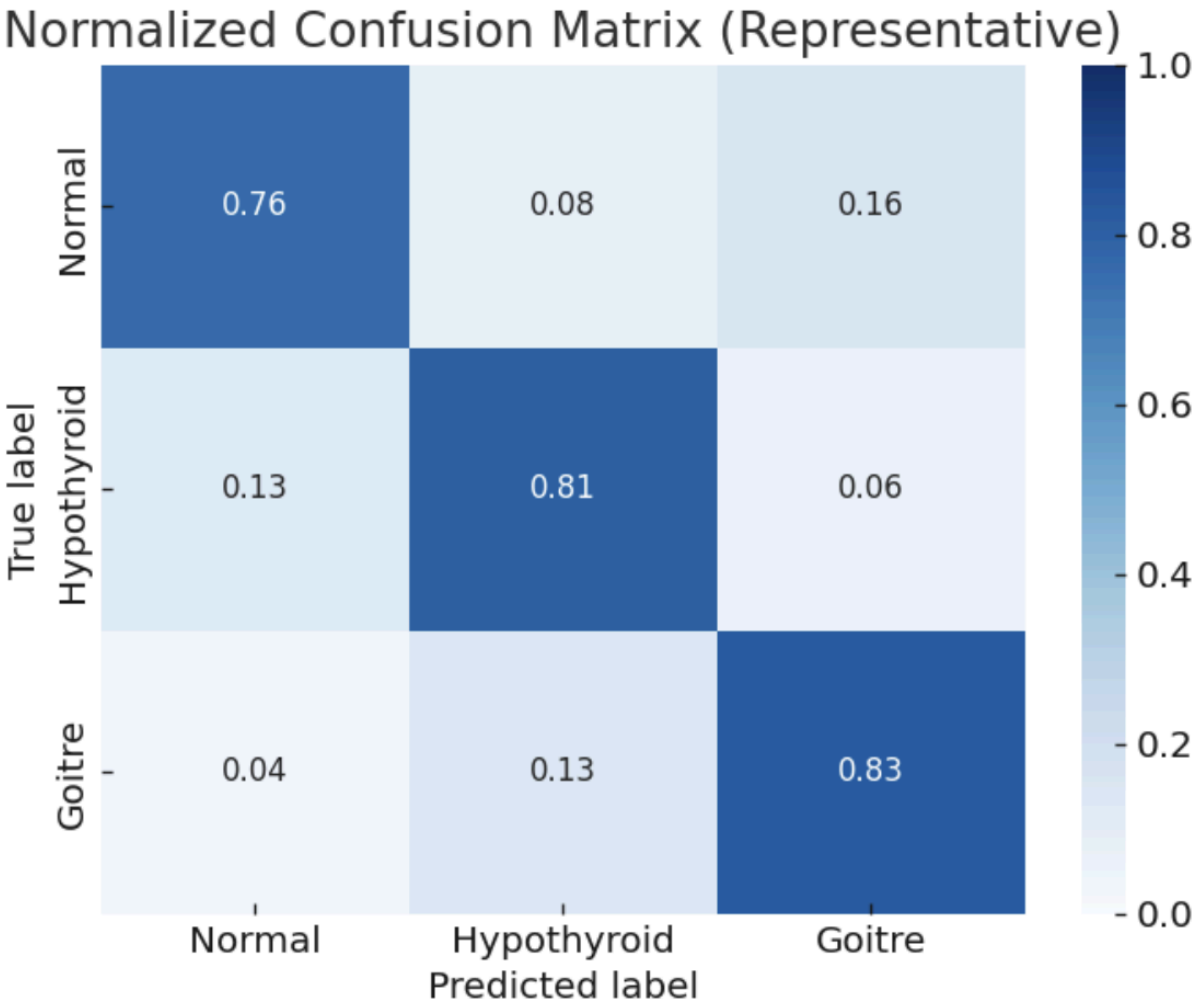

Fig. 4.4 Normalized Confusion Matrix

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Introduction

Artificial intelligence and machine learning are changing the face of healthcare, especially when it comes to early disease detection. Spotting thyroid disorders early can make a big difference—in how well treatments work and in keeping healthcare systems from getting overwhelmed. Thyroid issues are actually pretty common, but a lot of traditional tests depend on labs and special equipment that aren't available everywhere.

In this project, I set out to build a machine learning–based screening system that could flag thyroid disorders earlier, using information from a few different sources. The idea was to combine deep learning for analyzing neck images, a questionnaire that looks at symptoms, and a hormone-based machine learning model, bringing them all together in a hybrid prediction setup.

This chapter pulls together the main results, points out what the system achieved, talks through the roadblocks, and sketches where this work might go next.

5.2 Summary of the Proposed System

The system I put together uses a hybrid machine learning framework for thyroid screening. It doesn’t rely on just one thing—it looks at neck images, patient symptoms, and hormone levels to make better calls.

Everything runs in two steps. First, the system analyzes neck photos using a deep learning setup built on EfficientNetB0. Basically, it picks out patterns or problems in the neck that could signal thyroid trouble. Alongside that, patients answer a symptoms questionnaire. Their answers get converted to a weighted score—how likely it is that their symptoms match thyroid issues.

Then, the system fuses the image results with the symptom scores, creating a risk estimate based on both. If someone lands in a gray area—where the first answer isn’t clear—the system brings in hormone values (TSH, T3, T4) and runs a Random Forest classifier for a more precise verdict.

At the end, the system tells you what kind of thyroid problem it thinks is present, how risky it is, and gives advice about next steps or tests.

5.3 Achievements of the Project

This project showed a few important things. First, machine learning isn't just a buzzword—you really can use it to screen for thyroid issues before going to elaborate or expensive tests. By weaving together images, symptoms, and lab values, the system checks for thyroid disorders from more than one angle.

EfficientNetB0 turned out to be a solid choice for the neck image part. With transfer learning, it doesn't even need a giant pile of images to get decent results. Then, by adding in a symptom-based questionnaire, the system can pick up on thyroid problems that actually show up in what people feel, not just what doctors see.

Combining results from different sources—images, symptoms, hormones—really boosts the system's accuracy. It avoids mistakes that might slip in if you only looked at one thing. That second stage, where hormone values are brought in just for tricky cases, helps sharpen the results without making the process much more complicated or expensive.

All in all, the project is a good example of how AI can make healthcare screening more accessible and catch problems faster.

5.4 Limitations of the Proposed System

Of course, the system isn't perfect. One big challenge was the size of the image dataset. Deep learning loves lots of data, and thyroid images are hard to find—there just isn't a big public collection. That means the model might not do as well on totally new photos.

Image quality is another problem. Lighting, camera angles, messy backgrounds—all that can mess with the system's ability to spot patterns. There are also cases where thyroid disorders don't show up as physical changes in the neck, especially early on, so the image part alone won't always catch everything.

Symptoms in the questionnaire are tricky too. People experience things differently, and some symptoms pop up in other conditions, which makes the prediction fuzzier.

It's important to remember: this system isn't a final diagnosis tool. It's here to flag cases for further medical checks, not to replace a doctor's verdict.

5.5 Future Scope

There's a lot of room to grow here. Getting more and better image data would help the model generalize, so it works for more people and situations. Partnering with hospitals or clinics could bring in new data and make the tool stronger.

Adding ultrasound analysis could be a big win. Ultrasound images give a clearer look inside the thyroid and can catch things external photos miss. Mixing those with the other data could really sharpen diagnosis rates.

Explainable AI is another path—giving clear reasons for each prediction helps doctors trust the results. A mobile app could open up screening for everyone with a smartphone, letting people capture images, answer questions, and get rapid feedback.

Tying the system into electronic health records could make it even smarter, letting it consider medical history or medication use. And finally, working with real clinical teams to test and refine the system in the field would turn it from a project into a tool people can actually use.

5.6 Conclusion

To wrap up, the hybrid thyroid screening system I developed blends deep learning on neck images, patient-reported symptoms, and hormone test analysis to spot thyroid disorders early. It's a step toward using AI-driven tools to make early detection easier and more reliable.

While the system isn't without flaws, it shows that linking up different sources—visual, clinical, and lab data—can deliver a more complete and trustworthy assessment than any single method. With bigger datasets, improved models, and real-world testing, this kind of system could become a valuable ally for both healthcare pros and everyday users, boosting awareness and catching thyroid problems sooner.

Ultimately, the project highlights how machine learning can drive real changes in healthcare, making smart, accessible screening tools that lead to faster, better treatment for patients.

REFERENCES

[1] A. G. Unnikrishnan and U. V. Menon, “Thyroid disorders in India: An epidemiological perspective,” Indian Journal of Endocrinology and Metabolism, vol. 15, no. Suppl 2, pp. S78–S81, Jul. 2011.

[2] M. Tan and Q. V. Le, “EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks,” in Proc. Int. Conf. Mach. Learn. (ICML), Long Beach, CA, USA, 2019, pp. 6105–6114.

[3] S. Tyagi, A. Singh, and R. K. Singla, “Thyroid Disease Detection Using Supervised Machine Learning Techniques,” in Proc. 7th Int. Conf. on Comput. Methodologies and Commun. (ICCMC), Erode, India, 2023, pp. 945–950.

[4] W. Z. Billewicz, R. S. Chapman, J. Crooks, et al., “Statistical methods applied to the diagnosis of hypothyroidism,” Q. J. Med., vol. 38, no. 150, pp. 255–266, 1969.

[5] F. Pedregosa, et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[6] Y. I. Mir and S. Mittal, “Thyroid Disease Prediction Using Hybrid Machine Learning Techniques: An Effective Framework,” Int. J. Sci. Technol. Res., vol. 9, no. 2, pp. 2394–2399, 2020.

APPENDIX 1

Title: Clinical Symptom Questionnaire for Thyroid Pre-Screening

Instructions: Please indicate whether you are currently experiencing any of the following symptoms by selecting "Yes" or "No". Each response will be processed by our heuristic scoring module to calculate a preliminary clinical risk score.

Section A: General & Hypothyroidism Indicators

1. Do you experience persistent fatigue or unusually low energy levels? (Yes / No)

2. Do you have an increased sensitivity to cold temperatures compared to others? (Yes / No)

3. Have you experienced unexplained weight gain recently? (Yes / No)

4. Is your skin noticeably dry or rough? (Yes / No)

5. Do you frequently suffer from constipation? (Yes / No)

6. Have you noticed sudden depression or unexpected mood changes? (Yes / No)

7. Have you been told or noticed that you have a slow heart rate? (Yes / No)

8. Do you regularly experience unexplained muscle weakness? (Yes / No)

Section B: Hyperthyroidism Indicators

9. Do you frequently feel unusual anxiety or nervousness? (Yes / No)

10. Do you have a severe intolerance to heat? (Yes / No)

11. Have you experienced unexplained weight loss despite eating normally? (Yes / No)

12. Do you experience shaky hands or tremors? (Yes / No)

13. Have you noticed a rapid, fluttering, or irregular heartbeat (palpitations)? (Yes / No)

14. Do you experience excessive sweating even when inactive? (Yes / No)

15. Do you have frequent difficulty falling or staying asleep (insomnia)? (Yes / No)

Section C: Structural/Nodule Indicators

16. Can you feel or see a visible swelling at the base of your neck? (Yes / No)

17. Do you have difficulty or pain when swallowing? (Yes / No)

18. Has your voice become noticeably hoarse recently? (Yes / No)

19. Do you experience unexplained pain in your neck area? (Yes / No)

20. Do you occasionally have difficulty breathing or feel pressure in your throat? (Yes / No)

PROOF OF ACCEPTANCE OF RESEARCH PAPER

Gmail - Acceptance of Your Paper and Camera-Ready Submission Instructions 11/04/26, 12:45 AM

Gmail Lakshya Bhargava <lakshyabhargava11@gmail.com>

Acceptance of Your Paper and Camera-Ready Submission Instructions

1 message

Microsoft CMT <noreply@msr-cmt.org> 30 March 2026 at 11:33
To: Lakshya Bhargava <lakshyabhargava11@gmail.com>

Dear Author(s),

We are pleased to inform you that your paper entitled "A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening", submitted to FIRST INTERNATIONAL CONFERENCE ON NEXUS OF DIGITALIZATION, INTELLIGENCE AND APPLICATIONS, has been accepted for presentation and publication in IEEE conference proceedings, subject to the submission of the camera-ready version and final quality check by the internal screening committee.

Paper ID: 395
Paper Title: A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening

The reviewers comments are available in your Microsoft CMT login.

Kindly log in to your Microsoft CMT account and upload the camera-ready paper. Please ensure that the revised manuscript strictly follows the conference formatting guidelines and incorporates the reviewers' comments, wherever applicable. Please go through the reviewers comments, modify the paper and upload the paper. You are requested to upload camera ready paper on or before 15 April 2026.

Your paper strictly should be in IEEE format and Manuscript Templates for Conference Proceedings are given below:
https://www.ieee.org/conferences/publishing/templates

Please note that papers not complying with the IEEE format may not be included in the conference proceedings.

After uploading camera ready paper, the paper will be reviewed by the expert screening committee and final result will be conveyed to you within 15 days.

Warm regards,
Conference Review Committee
ICNDIA 2026
Department of Information Technology
Army Institute of Technology, Pune
http://www.aitpune.com
IEEE Conference Website: https://aitpune-icndia26.com/ or https://ieee-icndia26.com/

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our Privacy Statement.

Microsoft Corporation
One Microsoft Way
Redmond, WA 98052

PLAGIARISM REPORT

turnitin Page 2 of 12 - Integrity Overview Submission ID trn:oid:::1:3430567138

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **24** Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
5% Publications
3% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.